Prakhar Gupta

Full-Stack Developer | Open Source Contributor

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OPEN SOURCE

Sugar Labs (musicblocks-v4) – Open Source Contributor | [PR #646] **Jul 2026 – Present**

TypeScript, SVG Geometry

- Fixed SVG renderer geometry bugs by correcting stroke calculations and path-generation logic, resolving visual distortions across the MusicBlocks UI toolkit, merged by maintainer within 3 hours.

FaceGate-Mac – macOS Face-Auth App Locker | [Code] | [Web] | [Release] **Jun 2026 – Present**

Swift, AVFoundation, AppKit

- Contributed 3 merged PRs to a 300+ star macOS face-auth locker delivering security hardening (emergency kill toggle), concurrency fixes (camera race condition), and persistent overlay architecture with app-window mode — featured in v1.2.0 release.

EXPERIENCE

QMax QCom Optimizer – Freelance AI Engineer *(US Based Client)* **May 2026**

Architected an 8-agent ad-optimization platform replacing manual analysis across 4 Q-commerce apps.

Next.js, FastAPI, Prophet, Supabase, DuckDB, Claude, GPT-4o, DeepSeek

- Designed and delivered a full-stack multi-agent platform from scratch, reducing manual analysis workflows from days to seconds.
- Built a 3-tier LLM fallback chain with governed chatbot workflows, improving reliability during API/model failures while keeping raw data secure.
- Delivered a 14-page dashboard with sub-second queries on a 3-tier storage stack (Postgres + DuckDB), consolidating 4 siloed platforms into a single analytical view.

Home Tutor Platform – Freelance Full-Stack Developer *(Contract)* | [Live] **May 2026 – Jun 2026**

Delivered an SEO-optimized web platform with admin controls and a 3-channel lead capture system.

Next.js, NextAuth, Supabase, Resend, Tailwind CSS

- **Full-Stack Build:** Developed a 9-page public site and 12-section admin dashboard, enabling a non-technical client to manage SEO and content independently.
- **Backend Abstraction:** Engineered CRUD API powering 12 admin sections with role-based auth, eliminating massive code duplication.
- **Growth Architecture:** Deployed database-driven SEO metadata and dynamic sitemaps, generating direct leads through a unified WhatsApp + Email pipeline.

Bits & Bytes – Back-End Developer Intern | [Code] **Apr 2026 – May 2026**

Developed a full-stack Discord utility bot for a 1,000-member tech community, automating onboarding workflows, meeting scheduling with voice orchestration, audio transcription, and role-based access control.

Node.js, discord.js, SQLite, Notion API, Docker, FFmpeg

- **Platform Migration:** Redesigned the architecture with Docker and self-registering slash commands, eliminating deployment failures.
- **Centralized Auth & Storage:** Implemented role-based authorization gateways and hybrid SQLite+Notion storage, significantly reducing API latency.
- **Meeting System:** Built an end-to-end meeting scheduler with FFmpeg audio merging for live, timestamped transcripts.

PROJECTS

RailMind – Train Conflict Resolution Engine | [Code] | [Live] **Jun 2026**

Beam-search planning engine that reduces train priority conflicts across a 437km simulated corridor.

Python, FastAPI, NetworkX, React, WebSockets, SQLite

- **Planning Algorithm:** Implemented a beam-search engine that cuts passenger-delay costs by **10.07%** over FCFS dispatch with only **23ms** planner latency.
- **Robust Testing:** Built a disruption-injection system for 3 real-world failure types, rigorously validating the engine across 50 simulated corridor scenarios.
- **Explainability:** Developed a real-time WebSocket React dashboard with an interactive decision-tree visualizer, transforming black-box AI into transparent recommendations.
- **Real-World Validation:** Presented prototype to Khajuraho Railway Station Master and received positive feedback. [Post]

JanSamadhan – Civic Grievance Platform | [Live] | [Code] | [Demo] **Mar 2026**

Grievance routing platform piloted with a Delhi ward and winner at India Innovates '26.

FastAPI, Next.js, PostgreSQL/PostGIS, Redis, Gemini AI, WhatsApp Cloud API, GCP

- **AI Intake Pipeline:** Built a multilingual Gemini chatbot classifying complaints across 43 categories with auto-severity and authority routing at 3 paise per inference, paired with a full WhatsApp conversational flow supporting 10 Indian languages, image-first intake, and 20m spatial deduplication before ticket creation.
- **Database Architecture:** Designed an 8-table PostgreSQL schema with ENUM-enforced state machines, 28 Row-Level Security policies for department-isolated access, and PostGIS spatial indexes keeping ward-level queries under 100ms.
- **Computer Vision:** Deployed an ONNX-optimized YOLOv8 model at 20ms inference on GCP Cloud Run for automatic civic issue detection from CCTV feeds.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Swift, SQL, HTML/CSS, C++

Frontend / Backend: React, Next.js, Node.js, FastAPI, Tailwind CSS, REST APIs, WebSockets, NextAuth, discord.js

AI / ML: Multi-Agent Systems, LLM Orchestration, Gemini AI, Prophet, NetworkX.

Data / DevOps: PostgreSQL/PostGIS, SQLite, Redis, DuckDB, Docker, CI/CD, GCP, Railway/Render

Tools / Platforms: Git, FFmpeg, WhatsApp Cloud API, Supabase

ACHIEVEMENTS

India Innovates '26 – National Grand Finale Winner: JanSamadhan civic grievance platform [Post]

InnovateNSUT '26 – 1st Place: Leafline, a civic issue reporter [Post]

EDUCATION

Manipal University Jaipur – *Bachelor of Technology in Computer Science & Engineering* **Expected Graduation: May 2029**